

Rust Programming

Overview

Rust is a systems programming language focused on performance, memory safety, and concurrency. It provides low-level control similar to languages such as C and C++ while using a compile-time ownership system to prevent many memory errors.

Ownership

Rust's ownership system determines which part of a program owns each value. A value normally has one owner, and ownership can be moved between variables. References allow code to borrow values without taking ownership.

Borrowing

Rust allows immutable and mutable references with strict rules. These rules help prevent problems such as dangling pointers and data races while keeping memory management largely automatic.

Types and Enums

Rust provides structs for custom data structures and enums for values that can represent several possible variants. Pattern matching with `match` is commonly used to handle enum values safely.

Applications

Rust is used for command-line tools, operating systems, networking software, WebAssembly, embedded systems, and performance-sensitive backend services.